Aim :- Estimation of harðness of water by a standard solution of EDTA

Principle :--EDTA or its sodium salt forms soluble complexes with the hardness causing Ca^{2+} and Mg^{2+} ions.

$$Na_2H_2Y + Ca^{2+} = Na_2CaY + 2H^+$$
$$Na_2H_2Y + Mg^{2+} = Na_2MgY + 2H^+$$

When a pinch of solochrome black-T indicator is added to a solution containing Ca^{2+} and Mg^{2+} ions, at a pH 10, the solution turns wine red.To this solution, when EDTA solution is added as a titrant, the Ca^{2+} and Mg^{2+} ions are complexed. After sufficient volume of EDTA solution is added to complex all the Ca^{2+} and Mg^{2+} ions, the colour of the solution changes to pure blue, which is the end point of the titration. The hardness is expressed as parts by weight of $CaCO_3$ per million parts by weight of the water sample.

Procedure

This experiment involves the following steps :-

I. Preparation of a standard (0.1M) solution of $CaCO_3$ to standardize the EDTA solution.

II. Standardization of the EDTA solution by the standard solution of Ca^{2+}

III. Estimation of water by the standardized EDTA solution.

I. Preparation of a standard (0.1M) solution of $CaCO_3$:-

The molar mass of $CaCO_3 = 100$

So,100 mL 0.1M $CaCO_3$ solution $\equiv {}^{100}/_{1000} \times {}^{100}/_{10}$ g $CaCO_3$

$\equiv 1$ g $CaCO_3$

Table to determine the amount of $CaCO_3$ transferred :-

Initial weight of $CaCO_3$ & weighing bottle, (m_i)g	Final weight of $CaCO_3$ & weighing bottle, (m_f)g	Amount of $CaCO_3$ transferred **$m = (m_i - m_f)$g**

The transferred amount of $CaCO_3$ is dissolved in minimum volume of 1:1 HCl in a 100 mL volumetric flask and the remaining volume of the volumetric flask is filled with distilled water.

II Standardization of the EDTA solution :-

10 ml of the standard Ca^{2+} solution is pipetted out into a conical flask. 2 mL of $NH_3 - NH_4Cl$ buffer solution is added to it followed by 30 mg of solochrome black-T indicator, turning the solution wine red. This is titrated against the EDTA solution (taken in a burette) till the colour of the solution changes from wine red to pure blue, which is the end point . The titration is repeated to obtain two concordant readings.

Table to determine the volume of the EDTA solution (V_1) :-

No. of observation	Initial burette reading(mL)	Final burette reading(mL)	Volume of EDTA used(mL)	Concordant reading V_1 mL
1				
2				
3				

So, V_1 mL EDTA solution $\equiv$ 10 mL Ca^{2+} solution

$\equiv$ $^m/_{10}$ g $CaCO_3$

So, 1 mL EDTA solution $\equiv$ $^m/_{10V_1}$ g $CaCO_3$ --------(i)

III Estimation of hardness of the water sample:-

50 ml of the water sample is pipette out into a conical flask. 2 mL of $NH_3 - NH_4Cl$ buffer solution is added to it followed by 30 mg of solochrome black-T indicator, turning the solution wine red. This is titrated against the standardized EDTA solution (taken in a burette) till the colour of the solution changes from wine red to pure blue, which is the end point . The titration is repeated to obtain two concordant readings.

Table to determine the volume of the EDTA solution (V_2) :-

No. of observation	Initial burette reading(mL)	Final burette reading(mL)	Volume of EDTA used(mL)	Concordant reading V_2 mL
1				
2				
3				

So, 50 mL water sample $\equiv$ V_2 mL EDTA solution ---------(ii)

Calculation :-

(ii)=> So, 50 mL water sample $\equiv$ V_2 mL EDTA solution

or, 50 g water sample $\equiv$ $^m/_{10} \times V_2/V_1$ g $CaCO_3$ [Using (i)]

[Since density of water = 1 g/mL]

or, 10^6 g water sample $\equiv$ $^m/_{10} \times V_2/V_1 \times 10^6/_{50}$ g $CaCO_3$

$\equiv$ .... g $CaCO_3$

Conclusion :-

Estimated hardness of the given water sample = ... ppm $CaCO_3$

Aim of the experiment :- Estimation of Fe^{2+} by a standard solution of $KMnO_4$

This experiment involves the following steps:-

I. Preparation of a standard (0.1N) solution of oxalic acid to standardize the $KMnO_4$ solution.
II. Standardization of the $KMnO_4$ solution by the standard solution of oxalic acid.
III. Estimation of Fe^{2+} by the standardized $KMnO_4$ solution.

I. Preparation of a standard (0.1N) solution of oxalic acid :-

The equivalent mass of oxalic acid = 63

250 mL 0.1N oxalic acid solution $\equiv (63 \div 1000) \times (250 \div 10)$ g oxalic acid

$\equiv 1.575$ g oxalic acid

Table to determine the amount of oxalic acid transferred :-

Initial weight of oxalic acid & weighing bottle, (m_i)g	Final weight of oxalic acid & weighing bottle, (m_f)g	Amount of oxalic acid transferred **$m=(m_i - m_f)$g**	Strength of the solution **$(m \div 1.575) \times 0.1$(N)**

The solution is prepared by dissolving the oxalic acid in distilled water in a 250 ml volumetric flask and then making up the volume up to its mark .

II Standardization of $KMnO_4$ solution :-

20 ml of the standard oxalic acid is pipetted out into a conical flask. A test tube of dilute H_2SO_4 is added to it. The whole solution is diluted with distilled water to about 100 ml, warmed up to about 70°C and then titrated against the $KMnO_4$ solution (taken in a burette). The end point is attained when a faint pink colouration , persisting for about 20 seconds, is seen through the whole solution is obtained. The titration is repeated to obtain two concordant readings.

$$[\,C_2O_4^{2-} = 2CO_2 + 2e\,] \times 5$$
$$\underline{[MnO_4^- + 8H^+ + 5e = Mn^{2+} + 4H_2O] \times 2}$$
$$2\,MnO_4^- + 16H^+ + 5\,C_2O_4^{2-} = 2Mn^{2+} + 10\,CO_2 + 8H_2O$$

Table to determine the volume of $KMnO_4$ solution(V)

No. of observation	Initial burette reading(mL)	Final burette reading(mL)	Volume of $KMnO_4$ used(mL)	Concordant reading V mL
1				
2				
3				
4				

Let, S_1 = Strength of oxalic acid
V_1 = Volume of oxalic acid
Then the , $S = V_1 \times S_1 / V$
= ....(N)

III Estimation of Fe^{2+} :-

The given Fe^{2+} solution is diluted with distilled water up to the mark of the 100 ml volumetric flask in which it was supplied. 20 mL of this solution is pipetted out in to a conical flask and two test tubes of a mixture of dilute H_2SO_4 and H_3PO_4 is added to it. The whole mixture is diluted with distilled water to about 100 mL and then titrated against the standardized $KMnO_4$ solution. The end point is attained when a faint pink colouration ,persisting for about 20 seconds, is seen through the whole solution is obtained. The titration is repeated to obtain two concordant readings.

$$[Fe^{2+} = Fe^{3+} + e\ ] \times 5$$
$$[MnO_4^- + 8H^+ + 5e = Mn^{2+} + 4H_2O$$
$$MnO_4^- + 5Fe^{2+} + 8H^+ = 5Fe^{3+} + Mn^{2+} + 4H_2O$$

From above equation,

1 mol MnO_4^- ≡ 5 mol Fe^{2+}
or, 1 mol $KMnO_4$ ≡ 5 mol Fe
or, $^1/_5$ mol $KMnO_4$ ≡ 1 mol Fe
or, 1000 mL 1(N) $KMnO_4$ solution ≡ 56 g Fe
or, 1 mL 1(N) $KMnO_4$ solution ≡ 0.056 g Fe
or, V_2 mL S(N) $KMnO_4$ solution ≡ (V_2S X 0.056) g Fe
Hence the total estimated Fe^2 = 5 (V_2S X 0.056)g -----------(i)

Table to determine the volume of $KMnO_4$ solution(V_2)

No. of observation	Initial burette reading(mL)	Final burette reading(mL)	Volume of $KMnO_4$ used(mL)	Concordant reading V_2 mL
1				
2				
3				
4				

Calculation :-

Here, V_2 = Volume of $KMnO_4$ solution = ... mL
S = Strength of $KMnO_4$ solution = ...(N)

Using (i),

Estimated Fe^{2+} = 5 (V_2S X 0.056)g
= .... g

Conclusion :-

Estimated Fe^{2+} in the given solution = ... g

Estimation of Cu^{2+} by a standard solution of $Na_2S_2O_3$

This experiment involves the following steps:-

I. Preparation of a standard (0.05N) solution of $CuSO_4.5H_2O$
II. Standardization of the $Na_2S_2O_3$ solution by $CuSO_4.5H_2O$ solution.
III. Estimation of Cu^{2+} by the standardized $Na_2S_2O_3$ solution.

I. Preparation of a standard (0.05N) solution of $CuSO_4.5H_2O$

The molecular mass of $CuSO_4.5H_2O$ = 249.5 = Equivalent mass of $CuSO_4.5H_2O$
So,100 mL 0.05N $CuSO_4.5H_2O$ solution ≡ (249.5 ÷ 1000) x 100 x 0.05 g $CuSO_4.5H_2O$
≡ 1.2475 g $CuSO_4.5H_2O$

The initial weight of $CuSO_4.5H_2O$ & weighing bottle, m_1 =
The final weight of $CuSO_4.5H_2O$ & weighing bottle, m_2 =
So, the amount of $CuSO_4.5H_2O$ transferred, $m = (m_1 - m_2)g$ =

The solution is prepared by dissolving the $CuSO_4.5H_2O$ in 30 mL distilled water containing 2 drops of dil H_2SO_4 in a 100 ml volumetric flask and then making up the remaining volume up to its mark with distilled water.

II Standardization of $Na_2S_2O_3$solution :-

10 ml of the standard $CuSO_4.5H_2O$ solution is pipetted out into a conical flask. Dilute NH_4OH solution is added to it drop wise with constant shaking till a turbidity appears followed by dilute acetic acid, also drop wise and with constant shaking till that turbidity disappears. To this solution 30 mg of KI is added, the conical flask is immediately stoppered and kept in dark for 5 minutes. This solution, which is now dark brown, is titrated against the $Na_2S_2O_3$ solution (taken in a burette) till it turns pale yellow. At this stage, 2 ml of starch solution is added turning the solution bluish black and the titration is continued till the bluish black colour just disappears. This is the end point. The burette reading is noted and the whole process is repeated to obtain at least two concordant readings.

Reaction involved
(a) When KI is added to a solution containing Cu^{2+} the following reaction takes place.

$$2I^- = I_2 + 2e$$
$$Cu^{2+} + e = Cu^+]\, x2$$
$$2\,Cu^+ + 2I^- = Cu_2I_2$$

Adding, $$2Cu^{2+} + 4I^- = Cu_2I_2 + I_2$$

(b) When I_2 liberated in (a) is titrated against $Na_2S_2O_3$ solution the following reaction takes place.

$$2\,S_2O_3^{2-} = S_4O_6^{2-} + 2e$$
$$I_2 + 2e = 2I^-$$

Adding, $$2\,S_2O_3^{2-} + I_2 = S_4O_6^{2-} + 2I^-$$

Table to determine the volume of $Na_2S_2O_3$ solution (V_1) :-

No. of observation	Initial burette reading(mL)	Final burette reading(mL)	Volume of $Na_2S_2O_3$ used(mL)	Concordant reading V_1 (mL)
1				
2				
3				
4				

V_1 ml $Na_2S_2O_3$ solution $\equiv$ 10 ml of $CuSO_4.5H_2O$ sol^n
$\equiv$ (m/100) x 10 g $CuSO_4.5H_2O$
$\equiv$ m/10 g $CuSO_4.5H_2O$
So, 1 ml of $Na_2S_2O_3$ solution $\equiv$ (m/10) x $1/V_1$ g $CuSO_4.5H_2O$
$\equiv$ (63.5/ 249.5)x (m/10) x $1/V_1$ g of Cu^{2+} ------------- (i)
[Since, 249.5 g $CuSO_4.5H_2O \equiv$ 63.5 g Cu^{2+}]

{m is the mass of $CuSO_4.5H_2O$ dissolved to prepare the standard $CuSO_4.5H_2O$ sol.}

III Estimation of Cu^{2+} :-

The given Cu^{2+} solution is diluted with distilled water up to the mark of the 100 ml volumetric flask in which it was supplied. 10 mL of this solution is pipetted out in to a conical flask. Dilute NH_4OH solution is added to it drop wise with constant shaking till a turbidity appears followed by dilute acetic acid, also drop wise and with constant shaking till that turbidity disappears. To this solution 30 mg of KI is added, the conical flask is immediately stoppered and kept in dark for 5 minutes. This solution, which is now dark brown, is titrated against the standardized $Na_2S_2O_3$ solution (taken in a burette) till it turns pale yellow. At this stage 2 ml of starch solution is added turning the solution bluish black and the titration is continued till the bluish black colour just disappears. This is the end point. The burette reading is noted and the whole process is repeated to obtain at least two concordant readings.

Table to determine the volume of $Na_2S_2O_3$ solution (V_2) :-

No. of observation	Initial burette reading(mL)	Final burette reading(mL)	Volume of $Na_2S_2O_3$ used(mL)	Concordant reading V_2 (mL)
1				
2				
3				
4				

V_2 ml $Na_2S_2O_3$ solution ≡ 10 ml of supplied Cu^{2+}solution -------------------(ii)

Calculation :-

(ii) => 10 ml of supplied solution ≡ V_2 ml $Na_2S_2O_3$ solution

≡ (63.5/ 249.5)x (m/10) x V_2/V_1 g Cu^{2+}

So, 100 ml of the supplied solution ≡ (63.5/ 249.5) x (m/10) x V_2/V_1 x10 g Cu^{2+}

≡

Conclusion :-

Estimated amount of Cu^{2+} in the given solution = ------------- g

Aim: Determination of coefficient of viscosity of 5% glycerol solution at room temperature by Ostwald's Viscometer.

Requirements: Ostwald's viscometer, stop watch, pyknometer, graduated pipette, distilled water, 5% glycerol solution.

Procedure:

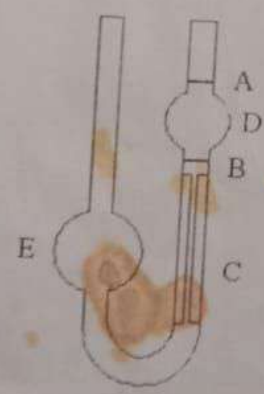
1. Thoroughly clean the viscometer with warm chromic acid and then rinse it several times with distilled water.
2. Attach a piece of clean rubber tubing to the top of the limb above A and clamp the viscometer vertically in a stand.
3. By means of a graduated pipette introduce 13 mL of distilled water into the arm E. With the help of the rubber tubing suck up the water in the capillary arm of the viscometer until the meniscus of the liquid is above the mark A. Allow the liquid to flow back by releasing the pressure. Start the stop watch when the meniscus passes the mark A and stop when it reaches the lower mark B. Record the time of flow of the distilled water from the mark A to B.Repeat the determination to obtain 2/3 readings which are in agreement to within 0.2-0.3 sec.
4. Empty the viscometer, rinse it with little of the 5% glycerol solution and clamp again into the stand vertically.
5. Fill the viscometer with 13 mL of the 5% glycerol solution and record the time of flow as before.
6. Determine the specific gravity of the 5% glycerol solution at room temperature with the help of the pyknometer.

Theory:

If h is the mean difference of the levels of the liquids in the two limbs during the flow and ρ_l is the density of the liquid, the force which resists this flow depends upon the dimensions of the capillary and viscosity of the liquid. Thus, the time of the flow of the liquid from the mark A to B for a given capillary is directly proportional to the driving force, i.e:

$$t_1 \alpha\ \eta_1/h\rho_1 g \quad \text{------------} \quad (1)$$

where η_l is the coefficient of the viscosity of the liquid.



Ostwald's Viscometer

If t_2 is the time of flow of the same volume of 5% glycerol solution from mark A to B in the viscometer, then

$$t_2 \,\alpha\, \eta_2/h\rho_2 g \quad \text{--------------} \quad (2)$$

where ρ_2 is the density of the 5% glycerol solution and η_2 is the co-efficient of viscosity of the 5% glycerol solution.

$$(1)\&(2) \Longrightarrow t_1/t_2 = \eta_1\rho_1/\eta_2\rho_2$$

$$\Longrightarrow \eta_1/\eta_2 = t_1\rho_1/t_2\rho_2$$

$$\Longrightarrow \eta_2 = \eta_1 t_2\rho_2/t_1\rho_1 \quad \text{-------------} \quad (3)$$

Thus, knowing the values of the coefficient of viscosity of the standard solution (i.e.: distilled water), the same of the test liquid (i.e. : 5% glycerol solution) can be determined.

Observations:

Room temperature =

(A) Determination of density:

Wt. of the empty sp. gravity bottle = w_1 g
Wt. of the sp. gravity bottle filled with distilled water = w_2 g
Wt. of the sp. gravity bottle filled with 5% Glycerol = w_3 g

So, $\rho_2 = w_3 - w_1/w_2 - w_1$

(B) Determination of t_1 & t_2:

Liquid	Flow time in seconds			
	(i)	(ii)	(iii)	Mean
Distilled water				t_1 =
5% Glycerol solution				t_2 =

Calculation:

$$\eta_2 = \eta_1 t_2\rho_2/t_1\rho_1$$

=

Conclusion:

Aim of the experiment: Determination of surface tension of a liquid at room temperature w.r.t water by drop number method using stalagmometer.

Requirements: Stalagmometer, small beaker, a piece of rubber tube, pyknometer, distilled water and experimental liquid.

Procedure:

1) Thoroughly clean the stalagmometer and the beaker several times with distilled water. Attach a piece of clean rubber tubing to the top of the stalagmometer.
2) Dip the flattened end of the stalagmometer in reference liquid (water) placed in a cleaned beaker and suck through the rubber tubing until the liquid level rises above the mark A.
3) Counting of drops is started when the water meniscus just reaches
   The upper mark A, and stopped when the meniscus just passes the lower mark B. Repeat to get three readings and take the mean value. Also record the number of divisions obtained for the formation of one drop.
4) Clean the stalagometer and fill it with liquid until it rises above the mark A and count the number of drops as before -------

Since the passage of the meniscus past these marks does not generally coincide with the falling of drops, a correction has to be applied in the total number of drops counted.

Suppose, the first drop formation commences at 5^{th} division below the upper mark and the last drop falls when the liquid meniscus passes 3^{rd} division below the lower mark. If the total number of drops counted are 55, then the corrected number of drops from a fixed volume of liquid between the two marks is equal to ----

55 + fraction of drop corresponding to 5 divisions - the fraction corresponding to 3 divisions.

Hence, surface tension is given by-

$$\gamma_1 / \gamma_2 = m_1 g F_1 / m_2 g F_2 = \upsilon_1 \rho_1 F_1 / \upsilon_2 \rho_2 F_2 \quad \text{------------------- (1)}$$

Where γ_1 & γ_2 are the surface tension of the reference liquid and the test liquids.

m_1 & m_2 are the masses of their drops.

F_1 & F_2 are the correction factors.

ρ_1 & ρ_2 are the densities of the reference liquid and the test liquids.

1

Aim of the experiment: Determination of surface tension of a liquid at room temperature by drop number method using stalagmometer.

The value F_1 and F_2 differ only slightly if the volumes of the drops of each of the liquids are not different then,

$$\gamma_1 / \gamma_2 = \upsilon_1\rho_1 / \upsilon_2\rho_2 \text{ ------------------- (2)}$$

If V is the volume of each liquid delivered and n_1 and n_2 are the number of drops counted for liquids 1 & 2 respectively, then

$$\gamma_1 / \gamma_2 = (V/n_1)\ \rho_1 / (V/n_2)\ \rho_2 = n_2\ \rho_1 / n_1\ \rho_2$$

$$\text{or, } \gamma_2 = (n_1\ \rho_2 / n_2\ \rho_1) \times \gamma_1 \text{ ------------------- (3)}$$

Observations:

(1) Room temperature =

(2) Determination of number of drops:

Liquid	Number of drops		Specific Gravity
	Drops counted	Mean	
Water		n_1	
Given liquid		n_2	

(3) Determination of density:

(i) Weight of the empty sp. gravity bottle $= W_1$ g

(ii) Weight of the sp. gravity bottle + water $= W_2$ g

(iii) Weight of the sp. gravity bottle + liquid $= W_3$ g

Therefore, specific gravity of the liquid is given by-

$$\rho_2 / \rho_1 = (W_3 - W_1) / (W_2 - W_1)$$

$$\text{or, } \rho_2 = \rho_1 \times (W_3 - W_1) / (W_2 - W_1)$$

Substitute the value of n_1, n_2, ρ_2 / ρ_1 and γ_1 (surface tension of the reference liquid water) at the same temperature in in eq. (3) and calculate the surface tension, γ_2 of the test liquid.

Conclusion:

NOTE: Unit of γ is Nm^{-1} (SI System) or $dynecm^{-1}$ (CGS system)

Surface tension of water at 25 ^0C = 7.197 x102 N/m

at 30 ^0C = 7.118 x102 N/m

Don't write

Aim: To determine the strength of the given HCl solution (strong acid) by titrating it against a standard 0.1(N) NaOH solution (strong alkali) conductometrically.

Apparatus Required: Conductivity bridge, conductivity cell, burette, beaker (250mL), Pipette (20mL)

Principle: When a strong acid (for example HCl) is titrated with a strong base (say NaOH) the following reaction takes place

$$(H^+ + Cl^-) + (Na^+ + OH^-) \longrightarrow Na^+ + Cl^- + H_2O$$

Thus, the highly conducting H^+ ions initially present in the solution are gradually replaced by low conducting Na^+ ions and therefore initially the conductivity value of the solution gradually decreases. However as soon as the solution becomes alkaline (after neutralization point) the solution becomes rich in more labile OH^- ions and hence the conductivity value of the solution increases again. So, at the equivalent point the conductivity is minimum.

Plot of conductance value of the solution as a function of the volume of NaOH produces a curve with two limbs and the intersection point of these two limbs provides the neutralization point of the titration.

Procedure:

1. Washed the conductivity cell thoroughly with distilled water and then rinsed the cell once with the solution whose conductivity is to be measured.

2. Pipetted out 10mL of the supplied acid solution of unknown strength in to a 200mL beaker and diluted the same with 100 mL of de-ionised water.

3. Filled the burette with the supplied base solution of known strength.

4. Dipped the conductivity cell in to the acid solution and the conductivity value is noted.

5. Added the NaOH solution from the burette carefully with a regular interval of 0.5 ml. After each addition the conductivity of the solution is noted. Initially, each addition of the alkali results in decrease of the conductivity of the solution. When after addition of few mL of the alkali solution the conductivity starts increasing, 6-7 more readings are taken to complete the experiment.

Table:

S. No.	Volume of NaOH added (cm^3)	Conductivity (siemen)

Calculations:

Volume of unknown acid pipetted = V_1 mL

Strength of unknown acid = S_1

Volume of base consumed = V_2 mL

Strength of base used = S_2

Therefore strength of unknown acid = $S_1 = V_2S_2/V_1$ (N) = ...(N)

Estimated amount of acid in the given sample = ...g/L

Conclusion: - Strength of the given acid is found to be ....(N) and ....g/L